Personal Notes

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1 [Shih and Tsai, 2004]

Discusses the selection bias that occurs when using a greedy (max gain) approach for building trees. They propose using a p-value based selection method. Where they show, using a simulation study, that the p-value based method for splitting results in a tree with least bias. In their simulation study they vary parameters such as missing entires and degree of dependece in the covariates. They conclude that using the p-value methods results in the least bias. Since the p-values is difficult to compute, they compare different approximation methods for calculating the p-value. The way they choose a split points is by computing the p-value of a split for each covariate and choose the covariate with smalles p-value as the split. Each p-value is calculated using a max test statistic. It is based on a ratio.

However, at a first read it seems that the selection issue they refer to only occurs when missing values are at play. Maybe I am wrong.

In the introduction they mention some references for where the Bonferonni adjustment can over-correct (Kass 1995, Scott and Knott 1976, Hawkins 1997).

Look into the references they provide for approximating p-values. It might be applicable to us since it looks very familiar.

2 [Yao and Davis, 1986]

They say that the heuristic approach of Hawkins 1977 is not entirely correct.

3 [Jensen and Schmill, 1997]

Discusses a similar approach to us. The idea is that, one needs to decide wheter or not to accept a split. This can be done by computing a score x for a split, where x can be accuracy or any other complex scoring rule. Then a split should be accepted if $x \geq x_c$, for some threshold x_c. In a greedy approach, one can choose x_c to be the score of the leaf without the split. x_c can be viewed as the gain.

The paper proposes an inference based method to determine wether to accept a split or not. However, they argue that when many covariates are at play, using a simple hypothesis test for a single covariate A is insufficient. Therefore, they propose the use of a Bonferoni correction to adjust for multiple testing. That is, they say that the real significance should be

$$\alpha_t = 1 - (1 - \alpha_i)^k$$

where, α_i is the significance for each (covariate) test and k is the number of covariates present at the node. This accounts for all possible tests.

I don't really get how they build a tree. However, they prune trees by setting some significance level α_t and checking if the significance of each *frontier* node $p \leq \alpha_t$, where p is calculated using Bonferoni correction. If so, keep that node otherwise prune it. Continue this procedure until all frontier nodes are within α_t. They call nodes, that have leaves as children frontier nodes.

They show that, they get superior models in terms of accuracy and complexity in most comparisons.

4 [Hawkins, 1977]

This paper talks about how the test statistic for change point is disitributed, in two cases, for known and unknown σ. They assume that the response is has normal error (with σ^2 variance).

In section 2, they speak breifly about the distribution of the ratio being larger than $\sigma^2\chi^2$ because of the maximizazation operation.

After the proof of thm 2 they provide an over estimate of the max distriubtion in with a scaled normal disctribution. Note that this is for the case when σ

5 [Fithian et al., 2017]

The **file drawer** example in the introduction is a good example on what selective inference is. It explains that, in face of selection, the true type-1 error rate is typically higher than the error rate for the simple test without taking into account selection.

They argue that instead of controlling the type-1 error by controlling the quantity

$$\mathbb{P}_{M,H_0}(\text{reject } H_0) \leq \alpha,$$

where, M is the model and H_0 is the null-hypothesis, they propose that the quantity that should be controlled is instead

$$\mathbb{P}_{M,H_0}(\text{reject } H_0 | (M, H_0) \text{ selected}) \leq \alpha.$$

They use an interesting metaphor relating to data splitting and selective inference. One can view data splitting in the following way. The first part where we select M, H_0 using data Y. This "happens before" the collection of the data used for inference. This is motivated by the fact that the 2 equations above are equivalent in when inference on M, H_0 is based on data independent of the selection process. Further, they state that this is the view they will have throughout the paper, where first we observe Y, then from Y we select M, H_0. Then we can treat $(Y|(M, H_0)selected)$ as "not yet observed".

In the lasso example, they state that controlling for selective inference is equivalent to conditioning on the region of space A_M such that, if $Y \in A_M$ then we will produce the same modell.

Definition 5.1 (Selective Control)**.** Let $\phi_q(y) \in [0, 1]$ represent the probability of rejecting H_0 if we have the outcome $Y = y$. Then we say that ϕ_q controls *selective type I error* at level α if

$$\mathbb{E}\left(\phi_q(Y)|A_q\right) \leq \alpha \text{ for all } F \in H_0(q).$$

Here, A_q is the set of events where the questions q is asked.

Definition 5.2 (Selective Power)**.** The *selective power function* is defined as

$$\text{Pow}\left(F|A_q\right) = \mathbb{E}\left(\phi_q(Y)|A_q\right)$$

Definition 5.3 (Selective confidence set)**.** We call $C(Y)$ a $1-\alpha$ *selective confidence set* if

$$\mathbb{P}_F(\theta(F) \in C(Y)|A_q) \geq 1 - \alpha.$$

From the above, do we really have a problem of selective inference? Since we are choosing a test statistic which is not based on a model but on the max, do we really suffer from modell selection? Further, can we build a tree and make inference on the whole tree, maybe building with Breiman's CCP and try to see if the difference between means is significant? Can we use the information that the test statistic we have found from Hawkins being based on likelihood ratio test can save us from selective inference?

6 [Hothorn et al., 2006]

Unbiased!

7 [Neufeld et al., 2022]

In their results, they calculate p-value in the following way.

8 [Zhang and Yu, 2005]

boost

9 [Bühlmann and Yu, 2003]

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